

Yuyuan Huang

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EDUCATION

Brown University, *Sc.B Computer Science and International and Public Affairs*, 4.0/4.0 GPA Providence, RI | **May 2028**
Relevant Courses: Software Engineering, Data Structures and Algorithms, Accessible and Inclusive Cybersecurity
The Winsor School, *High School Diploma*, 4.0/4.0 GPA Boston, MA | **May 2024**

SOFTWARE ENGINEERING EXPERIENCE

Medtronic, *Software Engineering Intern* Boston, MA | June 2025 – August 2025

- Built a real-time **graphical gaze tracking detector** for the HUGO surgical robot display using Python, JavaScript, QML, and C++, **improving debugging efficiency by 25% for 200+ engineers** in the Boston office.
- Incorporated UI/UX design principles to create an **intuitive and unobtrusive screen display** and ensure stable performance under surgical conditions.

FULL-STACK PROJECTS

C@bnet *React.js, TypeScript, HTML/CSS* | November 2025 – present

- Developing a **Chrome extension** to add integrated course ratings/warnings, a complex decision tree-based recommendation system, and degree pathway visualization to the current Courses@Brown website **used by 2000+ students and professors for course registration** at Brown University. Source code available upon request.

iMessages Wrapped *React.js, TypeScript, HTML/CSS* | December 2025

- Built a **full-stack animated analytics dashboard** for Apple iMessage group chat data using Python, Plotly, TypeScript with React, and Framer Motion. Designed interactive, sequentially animated data visualizations with SVG rendering, focusing on creative digital storytelling. Source code available upon request.

Illuminate *JavaScript, HTML/CSS, Python* | January 2025

- Worked with a team to prototype a web app that detects gender-biased language and suggests neutral adjectives using PyTorch, word2vec, and Hugging Face models.
- Incorporated feedback from 10+ users during Hack@Brown 2025 Showcase to refine UX and improve recommendation accuracy, **prioritizing features that enhanced usability**.

DebateBase *Python, SQL* | August 2023 – January 2024

- Developed a full-stack **user-centered educational web platform** to support 30+ underrepresented youth debaters in the Boston area with accessible **AI-driven learning tools**, advancing equity in public speaking education.

RESEARCH EXPERIENCE

Brown Human-Computer Interaction Lab, *Undergraduate Researcher* Providence, RI | September 2025 – present

- Leading frontend development and user research** for Cheeseburger Therapy, a CBT-based AI peer support platform focused on mental health accessibility and safety. Formulating research questions, engagement metrics, and analytical frameworks to assess risk, analyze user trust, and measure well-being in AI-guided mental health systems.
- Conducting qualitative user research to identify risks in AI-assisted healthcare applications, informing system design and regulation. Developing responsible AI guidelines for transparency and human oversight in clinical therapeutic settings.

Brown AI Research and Ethics Society, *Undergraduate Researcher* Providence, RI | October 2024 – September 2025

- Developed low-resource language models for Kurdish and Narragansett** using Python and HuggingFace, supporting linguistic equity for 1,800+ people in underserved linguistic communities across Rhode Island.
- Conducted research on bias mitigation and culturally responsive AI design to develop a written report on socially responsible technology development for natural language processing.

Harvard University, *Undergraduate Researcher* Boston, MA | June 2024 – August 2025

- Analyzed urban air quality disparities in disadvantaged Boston neighborhoods using Python, R, and Excel, synthesizing findings into presentations for **200+ community stakeholders** to support public health policy decisions.
- Built a **Python-based automated literature review tool integrating web scraping and generative AI to reduce manual search time on research databases by 25%**.

SKILLS

Programming Languages: TypeScript, JavaScript, HTML/CSS, Python, Java, Git, QML, C++

Technical Tools: React.js, Express, PostgreSQL, Node.js, Flask.js, Firebase, VS Code, Qt, PyTest, JUnit, Jest, Playwright

Design Tools: Figma, Miro, Adobe Suite, Microsoft Office Suite

Foreign Languages: Mandarin (native), Cantonese (native), Swedish (intermediate)

Additional Honors: National Center for Women and Information Technology (NCWIT) Aspirations in Computing Regional Affiliate Award, Massachusetts State Science and Engineering Fair 3rd place Award in Computer Science, National Girl Scout Gold Award, National Presidential Volunteer Service Gold Award, National Scholastic Art and Writing Awards Gold Medal, International Independent Schools Public Speaking Competition Qualifier